ALEX WAWERU

AI Engineer | Agentic AI Platforms, Applied ML & Secure Multi-Tenant Systems
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PROFESSIONAL SUMMARY

AI & Machine Learning Systems Engineer specializing in applied data science, machine learning, agentic AI, and production-scale AI platform engineering. Combines a strong foundation in statistical modelling and computational data science with hands-on expertise in engineering intelligent systems, from predictive and deep learning models to autonomous AI agents and scalable AI infrastructure.

Lead architect of **Nexus**, an AI Platform-as-a-Service (AI PaaS) at AICE Africa, providing unified access to independent AI/ML services through a graphical Agent Studio, SDKs, REST APIs, and Model Context Protocol (MCP). Experienced in engineering agent runtimes using LangGraph and LangChain, dynamic ReAct agents, layered memory architectures, retrieval systems, recommendation engines, forecasting, and computer vision.

Champions security-first AI architecture, incorporating PostgreSQL Row-Level Security (RLS) for tenant isolation, sensitive-data protection within agent workflows, gVisor-sandboxed code execution, and auditable agent actions. Holds a **BSc in Applied Statistics with Computing**, complemented by practical expertise in LLM engineering and fine-tuning, time-series analysis, econometric modelling, classical machine learning, and deep learning.

PROFESSIONAL EXPERIENCE

Senior AI Engineer
Jan 2026 – Present

AICE Africa (an Adanian Labs company) · *Nairobi, Kenya*

- Lead the architecture of AICE Nexus: agents, knowledge, crawler, Data Brain, recommendations and a sandbox, each usable alone or through MCP.
- Moved the agents from conditional routing to dynamic ReAct agents with tools, and from static to dynamic graphs, on a LangGraph runtime with a 24-slot middleware stack.
- Added long-term (semantic), episodic and in-context memory, gVisor sandboxing and an agent test harness.
- Built Agent Studio, which lets teams design and build agents graphically in Nexus, with skills.
- Built a hybrid recommendation engine, a custom web-search crawler and Data Brain, a data-source control layer with 10+ connectors including S3.
- Improved the knowledge base (RAG) with BM25 hybrid search, cross-encoder re-ranking and relevance scores from semantic distance, and built the customer and staff console.
- Postgres row-level security as the tenant boundary (per-service DB roles, raw-SQL isolation tests, four ADRs); RS256 project keys verified through JWKS with revocation, short-lived internal tokens, mutual-approval cross-tenant sharing, SSRF-guarded external MCP servers.
- Sensitive data stays out of agent context: sealed connection configs, PII-tokenised Parquet export, agents never see email addresses or biometrics, and generated Python runs in a no-network gVisor sandbox (p50 cold start ~0.87 s on an earlier build).

Junior AI Engineer
May 2025 – Dec 2025

AICE Africa (an Adanian Labs company) · *Nairobi, Kenya*

- Built the data analyst agent for a health research institute.

- Built agents with LangGraph, LangChain and FastAPI, MCP for tools, human-in-the-loop through LangGraph interrupts, short-term memory and Supabase.
- Built a RAG system in LangGraph, with linear nodes and conditional routers that hand off to subagents.
- Set up a Docker container sandbox for generated code, and model configuration.
- Worked on an onboarding agent system that completes forms.
- Cut the agent's fixed prompt by 67–79% with tool search.

Data Scientist (Freelance) — Aug 2020 – Present
Independent · *Remote*

- **Energy price forecasting & procurement platform (PJM wholesale market).** Built end to end: ingestion of PJM day-ahead/real-time LMPs, load forecasts, weather and EIA gas data; 72-hour direct multi-horizon LightGBM quantile forecasts (p10–p90) with horizon-aware buy signals; a VAR model on the energy, congestion and loss components of LMP; 36-month gas × heat-rate forward curves (Monte Carlo, rolling-origin backtests); and a weighted signal engine giving contract-aware (12/24/36-month) buy recommendations, customer hedge tracking and alerts. FastAPI + Celery, Supabase/Postgres with a pg_cron nightly pipeline, Kong gateway and a Next.js dashboard.
- **Contradiction detection (2024).** Fine-tuned Mixtral-8x7B-Instruct with QLoRA to flag contradictions in documents.
- **Audio transcription & speaker detection (H1 2024).** Split recordings into speaker-turn chunks with pyannote diarization, transcribed each chunk with Whisper, and used Mixtral 8x7B (Hugging Face, LangChain) to assemble a who-said-what dialogue.
- ML for sales platforms to improve resource utilisation; Power BI dashboards for service KPIs and revenue; econometric models of financial-market volatility and crises.

Data Scientist Intern — Jul 2024 – Dec 2024
KenGen · *Nairobi, Kenya*

- **Reconciliation automation.** Automated the energy-sector reconciliation workflow end to end: a manual multi-step computation became a single upload-and-download, with computation and visualisation automated and processing time cut to ~2 seconds. Built the proof of concept in R and the MVP in Python.
- Dash and Power BI KPI dashboards on MySQL; web-scraping tools that alert stakeholders to policy updates.

Data Scientist / AI Trainer — Mar 2024 – May 2024
AICE Africa · *Nairobi, Kenya (Remote)*

- Trained learners in Python, machine learning, evaluation, deployment, version control and data storytelling.

Data Analyst Intern — May 2022 – Jul 2022
SAMCHI Telecommunication · *Kenya*

- R reporting automation that cut manual effort by 80%; sales analysis and revenue-discrepancy investigations with finance.

SELECTED PROJECTS

- **Data Analyst Agent for APHRC (A-SEARCH HDSS)** *LangChain 1.0/LangGraph, Gemini, PostgreSQL* Sole author of the agent backend: data-engineer, data-analyst and ML-engineer subagents, ~100 MCP recipes (demographic indicators, 42 chart types, forecasting), hybrid RAG with reranking, a Dockerised session sandbox, token quotas and an admin dashboard. The model sees only schema and a 3-row preview.
- **Conversation Engine (multi-tenant AI CRM)** *FastAPI, LangChain, Supabase* Built on Nexus AI tools with agents designed in Agent Studio. Worked on the AI services which include: workflow-as-skill generation, multi-tenant RAG and crawler, PII hashing before the LLM, HMAC-authenticated MCP servers, an external-data connector (API/MySQL/Postgres), multi-provider LLM abstraction and a PPT/doc report generator.

- **Hybrid recommendation system (production)** *PyTorch, Gemini, pgvector* BPR-loss recommender with Gemini content embeddings and pgvector/HNSW lookup in production; the platform recommendation engine is a three-stage design (HNSW candidates, two-tower ranking, re-ranking) with tiered cold start and recall@10/NDCG@10 evaluation.
- **DataInViz agent skills (contributor)** Built the on-demand skills system (built-in and member-created), per-agent tool scoping and prompt-size optimisation for the platform's LLM controller.
- **Deepfake image detector (prototype)** *Swin Transformer, ZenML, MLflow* 0.970 accuracy / 0.994 AUC on 17k deduplicated images; a metadata-only leakage audit (floor 0.58) led to excluding two shortcut-prone sources.
- **GPT from scratch** *PyTorch* BPE tokenizer, dense and LLaMA-style (RoPE, RMSNorm, SwiGLU) models, trainer and evaluation harness; trained on 103 Kenya Gazette PDFs (val top-1 0.465, perplexity 25.7).
- **Sickle cell image classification (project)** *YOLOv8, YOLO12, Swin* Trained the YOLOv8, YOLO12 and Swin models; co-wrote the report. Swin reached 0.991 accuracy / 0.994 F1 on a single ~114-image test split; YOLOv8 F1 0.887; GradCAM, 5-fold CV.
- **Air-pollution forecasting (project)** *Vector Autoregression (VAR)* Multivariate time-series model in which each variable is forecast from its own past values and the past values of the other variables in the system.
- **K, AICE Africa assistant** Customer-facing conversational assistant built end to end, with persona enforcement.

TECHNICAL SKILLS

Languages	Python | SQL | R | TypeScript
Agentic AI	LangGraph | LangChain | MCP | Agent Protocol | ReAct agents | Agent memory (semantic, episodic) | Multi-agent workflows | Tool calling | Skills | Human-in-the-loop | Guardrails
LLMs & Retrieval	Gemini | OpenAI | Anthropic | Ollama / Mixtral | RAG | Hybrid search | BM25 + cross-encoder reranking | Embeddings | Docling | Fine-tuning (QLoRA, PEFT, TRL) | Whisper | pyannote | LLM evaluation | Transformer pretraining
Machine Learning	PyTorch | scikit-learn | LightGBM/XGBoost | CNN, ViT, Swin, YOLO | Recommender systems | Forecasting (VAR, quantile LightGBM, Monte Carlo) | Econometrics
Backend & APIs	FastAPI | REST | Microservices | Celery | SDK & API design | Pytest | Next.js / React
Databases	PostgreSQL (RLS, pgvector/HNSW) | DuckDB | MySQL | Redis | Supabase
Security	Tenant isolation | RLS | JWT/JWKS | gVisor sandboxing | SSRF defence | Audit logs | Data sovereignty
Cloud & DevOps	Docker | GitHub Actions | Google Cloud | OpenTelemetry | Grafana/Prometheus | Self-hosting
Analytics & BI	Power BI | Dash | Pandas/NumPy

EDUCATION

BSc Applied Statistics with Computing | Moi University, Kenya 2023